Changkai Chen

List of Publications

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Peer-reviewed publications

Six first-author peer-reviewed articles and two first-author preprints.

[1] **Changkai Chen**, Zhoujian Cao, and Jiliang Jing. Efficient and stable computation of gravitational-wave fluxes from generic Kerr orbits via a unified Heun-function framework. Phys. Rev. D 114, 063002 (2026).

[2] **Changkai Chen**, Jiliang Jing, Zhoujian Cao, and Mengjie Wang. Complete quasinormal modes of type-D black holes. Phys. Rev. D 112, 103036 (2025).

[3] **Changkai Chen** and Jiliang Jing. Gravitational wave fluxes on generic orbits in near-extreme Kerr spacetime: Higher spin and large eccentricity. Sci. China Phys. Mech. Astron. 67, 110411 (2024).

[4] **Changkai Chen** and Jiliang Jing. Radiation fluxes of gravitational, electromagnetic, and scalar perturbations in type-D black holes: an exact approach. JCAP 11 (2023) 070.

[5] **Changkai Chen**, Xiaohua Zhang, and Zhang Liu. A high-order compact finite difference scheme and precise integration method based on modified Hopf-Cole transformation for numerical simulation of n-dimensional Burgers' system. Appl. Math. Comput. 372, 125009 (2020).

[6] **Changkai Chen**, Xiaohua Zhang, Zhang Liu, and Yage Zhang. A new high-order compact finite difference scheme based on precise integration method for the numerical simulation of parabolic equations. Adv. Differ. Equ. 2020, 15 (2020).

Preprints

[7] **Changkai Chen**, Xiaohua Zhang, Zhoujian Cao, Jiliang Jing, and Sheng Long. Complete total-transmission modes of Kerr black holes. arXiv:2609.39609 [gr-qc] (2026), preprint.

[8] **Changkai Chen**, Xiaohua Zhang, Zhoujian Cao, and Jiliang Jing. Complete quasinormal modes of Type-D black holes. II. Extremal Kerr black holes. arXiv:2610.39095 [gr-qc] (2026), preprint.